

# MIDTERM EXAM—SOLUTIONS

ECO403/503—INTRODUCTION TO CONTRACT THEORY & AUCTIONS

SEPTEMBER 30, 2025

## Problem 1: Pokémon Duel with Spillovers

Our two TAs, Sultan and Bagdat, duel for a rare shiny Pokémon card worth  $W > 0$ . Each chooses non-negative effort  $x_i$  (deck building, trading, scrimming). The probability that player  $i$  wins the duel is given by

$$p_i(x_i, x_j) = \begin{cases} \frac{x_i}{x_i + x_j} & \text{if } x_i + x_j \neq 0, \\ 1/2 & \text{if } x_i + x_j = 0. \end{cases}$$

Correspondingly, the probability that  $i$  loses the duel is given by  $1 - p_i(x_i, x_j)$ .

However, in the world of Pokémon dueling, battles have spillovers: even if you lose, you might still gain something from your rival's effort, e.g., learning new strategies, increasing your reputation in the circuit, picking up booster packs, etc.

Formally, the payoff of player  $i$  is:

$$u_i(x_i, x_j) = \begin{cases} W - x_i + \beta_1 x_j & \text{if } i \text{ wins,} \\ -x_i + \beta_2 x_j & \text{if } i \text{ loses,} \end{cases}$$

where  $\beta_1 \geq 0$  and  $\beta_2 \geq 0$  are parameters which capture the extent of Pokémon-card battle spillovers.

- 1.0.1. Write down the expected payoff of player  $i$ .
- 1.0.2. Compute the player  $i$ 's best response function. State a sufficient condition on the parameters under which the maximization problem of player  $i$  is strictly concave (and thus has a unique global maximizer).
- 1.0.3. Derive the symmetric Nash equilibrium.
- 1.0.4. How do the equilibrium efforts depend on the parameters  $\beta_1$  and  $\beta_2$ ? *Hint: think about whether equilibrium effort rises when the benefit from winning spillovers increases ( $\beta_1$ ), and whether it falls when the loser's spillover benefit ( $\beta_2$ ) increases.*

### Solution

- 1.0.1. Let  $\pi_i(x_i, x_j) = E[u_i(x_i, x_j)]$  denote the expected utility of player  $i$ . When  $x_i + x_j \neq 0$ , the expected utility is  $p_i(x_i, x_j)(W + \beta_1 x_j) + (1 - p_i(x_i, x_j))(\beta_2 x_j) - x_i$ . Similarly, when  $x_i + x_j = 0$ , the expected utility is  $W/2$ . After simplifying we get the following expression for the expected payoff:

$$\pi_i(x_i, x_j) = \begin{cases} \beta_2 x_j + p_i(x_i, x_j)A(x_j) - x_i & \text{if } x_i + x_j \neq 0, \\ \frac{W}{2} & \text{if } x_i + x_j = 0, \end{cases}$$

where  $A(x_j) = W + (\beta_1 - \beta_2)x_j$ .

- 1.0.2. Let  $p_{i,1}(x_i, x_j) = \frac{\partial p_i(x_i, x_j)}{\partial x_i}$  and  $p_{i,11}(x_i, x_j) = \frac{\partial^2 p_i(x_i, x_j)}{\partial x_i^2}$  denote, respectively, the first and second partial derivatives of  $p_i$  with respect to  $x_i$ . For  $p_i(x_i, x_j) = \frac{x_i}{x_i + x_j}$  (defined when  $x_i + x_j \neq 0$ ), these are

$$p_{i,1}(x_i, x_j) = \frac{x_j}{(x_i + x_j)^2}, \quad p_{i,11}(x_i, x_j) = -\frac{2x_j}{(x_i + x_j)^3}.$$

Differentiating the expected utility, for  $x_j > 0$ , we obtain:

$$\frac{\partial \pi_i}{\partial x_i} = p_{i,1}A(x_j) - 1, \quad \frac{\partial^2 \pi_i}{\partial x_i^2} = p_{i,11}A(x_j).$$

A sufficient condition for strict concavity, when  $x_j > 0$ , is  $A(x_j) > 0$ . A parameter-only sufficient condition that guarantees this for all  $x_j > 0$  is  $\beta_1 \geq \beta_2$ .

Assume  $\beta_1 \geq \beta_2$ , so that  $A(x_j) > 0$  for  $x_j > 0$ . For  $x_j > 0$ , the interior FOC

$$\frac{\partial \pi_i}{\partial x_i} = p_{i,1}(x_i^*, x_j) A(x_j) - 1 = \frac{x_j}{(x_i^* + x_j)^2} A(x_j) - 1 = 0$$

yields

$$x_i^* = \sqrt{A(x_j) x_j} - x_j.$$

This candidate satisfies  $x_i^*(x_j) \geq 0$  if and only if  $A(x_j) \geq x_j$ .

For  $x_j = 0$ , we have

$$\pi_i(x_i, 0) = \begin{cases} W - x_i & \text{if } x_i > 0, \\ \frac{W}{2} & \text{if } x_i = 0. \end{cases}$$

Hence the expected payoff of  $W$  is approached as  $x_i$  approaches 0 from above, but this value cannot be attained by any  $x_i > 0$ , so there is no best response at  $x_j = 0$ .<sup>a</sup>

Combining all cases, under  $\beta_1 \geq \beta_2$ ,

$$\text{BR}_i(x_j) = \begin{cases} \sqrt{A(x_j) x_j} - x_j & \text{if } x_j > 0 \text{ and } A(x_j) \geq x_j, \\ 0 & \text{if } x_j > 0 \text{ and } A(x_j) < x_j, \\ \emptyset & \text{if } x_j = 0. \end{cases}$$

- 1.0.3. At a symmetric interior equilibrium we set  $x_i = x_j = x^* > 0$ . Then using the best response function we computed above, we must have that  $x^* = \text{BR}_i(x^*)$ . Therefore:

$$x^* = \sqrt{A(x^*) x^*} - x^* \iff 4x^* = W + (\beta_1 - \beta_2) x^*,$$

which yields

$$x^* = \frac{W}{4 - (\beta_1 - \beta_2)}.$$

Since  $W > 0$  and we assumed  $\beta_1 \geq \beta_2$ , a unique symmetric interior equilibrium exists if and only if

$$\beta_1 - \beta_2 < 4,$$

in which case  $x^* > 0$ . Note  $A(x^*) = 4x^* > x^*$ , so the equilibrium indeed satisfies the appropriate condition in the best response. If  $\beta_1 - \beta_2 \geq 4$ , no symmetric interior solution exists;  $(0, 0)$  is not an equilibrium because there is no best response at  $x_i = x_j = 0$ .

- 1.0.4. As the winner's spillover benefit  $\beta_1$  increases, each player's marginal gain from winning rises, which strengthens incentives to exert effort. Hence equilibrium effort  $x^*$  increases with  $\beta_1$ . Conversely, when the loser's spillover benefit  $\beta_2$  increases, losing becomes relatively less costly, reducing the incentive to compete aggressively. As a result, equilibrium effort decreases with  $\beta_2$ . Intuitively, higher  $\beta_1$  raises the prize's effective value, while higher  $\beta_2$  introduces more "free-riding" by softening the sting of losing.

Although this example was phrased within a humorous setting involving a Pokémon duel between our TAs, it can also be interpreted as the reduced form of an R&D contest in which property rights are weakly protected. In such environments, losing firms may still enjoy large technological spillovers from the winner's innovations (a high  $\beta_2$ ), which reduces the effective cost of losing and thus discourages effort. The result is strong incentives to free ride on the effort of others.

<sup>a</sup>This is analogous to the Bertrand price competition game.

## Problem 2: The Guessing Game Experiment

Fifty students are recruited for an important experiment designed by Prof. Alejandro, who somehow secured a grant for this and managed to bypass human-ethical experimental restrictions in the process.

Each student writes one integer between 0 and 100 on a slip of paper, alongside their name. Prof. Alejandro collects all the slips and calculates the average  $x$  of all the numbers written by the students. The

winner of the game is the student (or students) who wrote down the number that is closest to  $\frac{2}{3}x$ . The winners equally divide the prize of \$1000 USD between them.

2.0.1. Describe this game as a static game in normal form: specify the players, their action sets, and their payoff functions.

2.0.2. Find all Nash equilibria of the game.

2.0.3. If you were one of the students, what number would you write on your slip, and why?

### Solution

2.0.1. The strategic form of the game in this task is given by  $N = \{1, \dots, 50\}$ , a set of players;  $A_i = \{0, 1, \dots, 100\}$  an action set for every player  $i \in N$ ; and if we let

$$x(a) = \frac{\sum_{i=1}^{50} a_i}{50}, \quad W(a) = \arg \min_{j \in N} \left| a_j - \frac{2}{3}x(a) \right|,$$

where  $a = (a_i)_{i \in N}$ ,  $x(a)$  is the average, and  $W(a)$  is the winner set, then

$$u_i(a) = \begin{cases} \frac{1000}{|W(a)|} & \text{if } i \in W(a), \\ 0 & \text{otherwise.} \end{cases}$$

where  $|W(a)|$  denotes the size of the winner set.

2.0.2. The following is an intuitive argument to arrive at an educated guess of what the equilibria must be, and it is not required for a complete answer. By iteratively eliminating weakly dominated strategies, we can see that any number above 67 can never be optimal, since the target  $\frac{2}{3}x$  cannot be that high. Repeating this argument we obtain that the surviving actions are the lowest integers, suggesting that any equilibrium should involve only 0's or only 1's. Guided by this intuition, we now prove that the only Nash equilibria are the two symmetric profiles  $(1, \dots, 1)$  and  $(0, \dots, 0)$ . If the action profile is  $(1, \dots, 1)$ , then  $x(1, \dots, 1) = 1$  in which case  $\frac{2}{3}x = \frac{2}{3}$ , so each player gets  $\frac{1000}{50}$ , whereas by unilaterally deviating to any other action a player would get zero. On the other hand, if the action profile is  $(0, \dots, 0)$ , then  $x(0, \dots, 0) = 0$  in which case  $\frac{2}{3}x = 0$ , so each player gets  $\frac{1000}{50}$ , whereas by unilaterally deviating to any other action a player would get zero.

We now proceed to establish that the above two profiles are the only Nash equilibria. In order to do that, we will consider three classes of profiles. Firstly, consider the profile in which all players choose the same number  $k$ , with  $k > 1$ , i.e., the profile  $(k, \dots, k)$ . The payoff to each player under the profile is  $\frac{1000}{50}$ , whereas by deviating to  $k - 1$  a player can get 1000.<sup>a</sup>

We next consider the action profile such that each player writes one of two different numbers, where each number is written at least once, i.e.,  $a_i \in \{w, z\}$  for all  $i \in N$  and for some numbers  $w, z$  in  $\{0, 1, \dots, 100\}$ , with  $w \neq z$ . Without loss of generality, assume  $w < z$ . Let  $N_w$  and  $N_z$  denote the sets of players that write  $w$  and  $z$  respectively, with sizes  $n_w$  and  $n_z$ . Notice that  $n_w + n_z = 50$ , and therefore

$$x = \frac{n_w w + n_z z}{50} = \frac{n_w w + n_z z}{n_w + n_z}.$$

We first consider the case in which  $|w - \frac{2}{3}x| = |z - \frac{2}{3}x|$ . This equation implies that

$$\frac{2}{3}x = \frac{w + z}{2} \iff \frac{2}{3} \frac{n_w w + n_z z}{50} = \frac{w + z}{2},$$

which implies that  $n_w \neq n_z$ , and together with  $w < z$ , we obtain  $n_w < n_z$ . In this profile, all players obtain  $\frac{1000}{50}$ . Then any player  $i \in N_z$  can deviate to  $w$  and still win; after this deviation the winner set size is exactly  $n_w + 1$  players choosing  $w$ , so the deviating player's payoff increases from  $\frac{1000}{50}$  to  $\frac{1000}{n_w + 1}$ .

On the other hand, now suppose, without loss of generality, that  $|w - \frac{2}{3}x| < |z - \frac{2}{3}x|$ . Then each player  $i \in N_z$  receives zero, and he gets a positive payoff by deviating to  $w$ .

Finally, consider now the profile  $a = (a_1, \dots, a_{50})$  where there are at least three different numbers. This profile cannot be an equilibrium. Building on the intuition from the previous case, if there are

at least three different numbers, then there must be at least one player whose payoff under  $a$  is zero. If any such player deviates, the average changes by at most  $\frac{100}{50}$  and thus the target shifts by at most  $\frac{2}{3} \frac{100}{50} \approx 1.33$ . Therefore, such a player can deviate to a number under which he wins the game.

- 2.0.3. Any reasonable line of reasoning would be acceptable in this case. If there is common knowledge of rationality, then playing 0 is a good choice, since it survives the iterative elimination of dominated strategies. If, on the other hand, one suspects that not all players are perfectly rational, the answer becomes trickier, as it depends on one's beliefs about the rationality and computational ability of the other participants. For example, if there is a "troll" player who simply wants to watch the world burn and writes 100, then 0 may no longer be a good strategy. Similarly, if some participants are, to put it bluntly, "stupid" in the strategic sense, unable or unwilling to reason through even one or two steps of the logic, then higher numbers will appear. Such players, together with those attracted to focal numbers like 50 or 100 simply because they look "natural," will drive the average upward and alter best responses accordingly. But rather than dismiss the theoretical prediction, this is precisely where the model becomes useful: it provides a structured way to test how far behavior departs from full rationality and common knowledge of rationality and why. To properly account for trolls, stupid players, and focal-point seekers alike, as well as departures from common knowledge of rationality, we need to introduce beliefs about others' reasoning processes and formalize them as *types*, which is exactly what a Bayesian game allows us to do.

<sup>a</sup>Make sure to do the simple algebra that validates this claim!

### Problem 3: The Coordinated TA-Attack Problem

Two divisions of the **CAT-TA Army**, with Akmaral leading the left wing and Sezim commanding the right wing, are secretly plotting a bold maneuver: to jump on their Head TA, Abuzer, in the NU atrium. If both strike at the same time, the rebellion succeeds, and Abuzer will finally let them do the specific tutorials they want. But if only one of them jumps while the other hesitates, the lone attacker will be captured and sentenced to Abuzer's ultimate punishment: being sent to hang out with the business administration bros.

Akmaral suggests a coordinated jump tomorrow at **10:00 AM sharp**, while Abuzer is going to one of his lectures. But neither she nor Sezim will dare to attack Abuzer unless they are absolutely sure the other is also attacking. After all, nobody wants to endure the torture of hanging out with the business bros.

Since Abuzer controls all the Telegram channels, the only way Akmaral and Sezim can communicate is through their friend Nozima, who acts as a courier for the messages. Nozima takes 10 minutes to get from one TA to the other. Abuzer usually patrols the courier routes, which also exposes Nozima to capture and the same torturous punishment of being sent to hang out with the business bros. Luckily, tonight Abuzer is busy preparing his PhD applications, so he is not patrolling the courier route.

**Question:** How long will it take before Akmaral and Sezim can actually coordinate the jump on their Head TA, Abuzer, inside NU? Justify your answer.

#### Solution

In order for Akmaral and Sezim to be absolutely certain that they will both jump on Abuzer at the same time, they must have *common knowledge of the attack*, meaning that each knows the plan, knows that the other knows, knows that the other knows that she knows, and so on *ad infinitum*.

Although Nozima can carry messages between them, this process can never create true common knowledge. Whenever Nozima is on her way from Akmaral to Sezim, Akmaral hesitates because she cannot be sure that Sezim has received the message. Even after Nozima delivers it, Sezim will not jump either, since she cannot be sure that Nozima has returned to confirm that Akmaral knows the message was delivered.

This chain of uncertainty continues forever. Each new message only adds one more layer of "I know that you know," but never achieves the infinite hierarchy required for common knowledge of the attack. Therefore, despite Nozima's heroic efforts, Akmaral and Sezim can never coordinate the jump on their Head TA, Abuzer, in finite time.

## Problem 4: Aumann's Agreement Theorem

Consider the following version of Aumann's Agreement Theorem.

**Theorem.** Let  $(N, \Omega, (\mathcal{P}_i)_{i \in N}, \mu)$  be a finite belief space with common prior  $\mu$  such that  $\mu(\omega) > 0$  for all  $\omega \in \Omega$ . Suppose that  $N = \{1, 2\}$ . Let  $A \subseteq \Omega$  be an event and let  $\omega \in \Omega$  be a state of the world. If the event "Player 1 ascribes probability  $q_1$  to  $A$ " is common knowledge, and the event "Player 2 ascribes probability  $q_2$  to  $A$ " is also common knowledge in  $\omega$ , then  $q_1 = q_2$ .

Below, I outline an argument that attempts to prove this result in a different way than the one we saw in class. The argument is summarized in 6 steps, and certain details are missing. In particular, each step contains an underlined statement that needs verification. Either prove these statements, or give a precise reason(s) why you would expect them to be false.

- (i) Let  $C$  be the connected component of  $\omega$  in the graph corresponding to the belief space above. It follows that  $C$  is common knowledge at  $\omega$ .
- (ii)  $C = \bigcup_j P_1^j$ , where  $P_1^j \in \mathcal{P}_1$  for each  $j$ .
- (iii)  $\mu(P_1^j) > 0$  for every  $j$ , and therefore  $\mu(C) > 0$ .
- (iv) The fact that Player 1 ascribes probability  $q_1$  to the event  $A$  is common knowledge in  $\omega$ .  
Then for each  $j$ ,  $\mu(A \mid P_1^j) = \frac{\mu(A \cap P_1^j)}{\mu(P_1^j)} = q_1$ .
- (v) It follows that  $\mu(A \cap C) = q_1 \mu(C)$ .
- (vi) We similarly derive that  $\mu(A \cap C) = q_2 \mu(C)$ . Since  $\mu(C) > 0$ , the previous steps imply  $q_1 = q_2$ .

### Solution

- (i) This is true and follows from the characterization theorem 3.22 in the slides.
- (ii) This is true but needs a small amendment for precision. From the previous statement, we know that  $C \in \Sigma_C = \Sigma_1 \cap \Sigma_2$ . In particular,  $C$  must be formed from the atoms (propositions 3.14 and 3.15) of  $\Sigma_1$ , which are precisely the partition elements of  $\mathcal{P}_1$ . Therefore, if we enumerate the elements of  $\mathcal{P}_1$  as  $P_1 = \{P_1^j\}_{j \in J}$ , where  $J$  is a finite index set<sup>a</sup>, then there exists  $J' \subseteq J$  such that  $C = \bigcup_{j \in J'} P_1^j$ .
- (iii) This is true and follows from the assumption that  $\mu(\omega) > 0$  for all  $\omega \in \Omega$ .
- (iv) This is true but requires a more complete argument. From the first statement it follows that the event  $\{\omega' \in \Omega : \mu(A \mid P_1(\omega')) = q_1\}$  contains the event  $C$  (this is precisely theorem 3.22). Since  $C = \bigcup_{j \in J'} P_1^j$ , we have for each  $j \in J'$  that

$$\mu(A \mid P_1^j) = \frac{\mu(A \cap P_1^j)}{\mu(P_1^j)} = q_1.$$

- (v) This is true and follows from the previous observations:

$$\mu(A \cap C) = \mu\left(A \cap \bigcup_{j \in J'} P_1^j\right) = \mu\left(\bigcup_{j \in J'} (A \cap P_1^j)\right) = \sum_{j \in J'} \mu(A \cap P_1^j) = \sum_{j \in J'} q_1 \mu(P_1^j) = q_1 \mu(C),$$

where the first equality follows from (ii), the second from standard set operations, the third from the disjointness of the partition elements, the fourth from (iv), and the last from the additivity of  $\mu$ .

- (vi) This is true, because the same argument from (ii) to (v) applies *mutatis mutandis* to player 2.

<sup>a</sup>A finite subset of the natural numbers, which we can assume has 1 as its smallest element.